

Can World Models Optimize Customer Journeys and Lifetime Value?

A diagnostic, simulation-based framework for sequential marketing optimization

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Abstract

World models—learned simulators of customer dynamics—offer a natural mechanism for optimizing customer journeys and customer lifetime value (CLV), where marketing actions have delayed effects through engagement, fatigue, and churn. However, learning a simulator from operational logs is challenging because action assignment is often confounded by targeting rules, and planning can amplify model error over longer horizons. We propose a diagnostic, simulation-based framework that separates *training* (learning a world model from logged trajectories) from *grading* (evaluating policies in an independent ground-truth simulator), enabling direct measurement of a *reality gap* between simulated and realized policy value. In Level-0 experiments, we observe a regime-dependent reality gap: models trained on randomized logs remain comparatively conservative on the analyzed subset, while models trained on confounded logs become increasingly optimistic at longer horizons, overestimating achievable value.

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1 Introduction

Marketing optimization has progressively moved from descriptive analytics toward prediction, causal targeting, and sequential decision-making. Customer journeys are multi-touchpoint and dynamic: interventions taken today influence future response. This motivates viewing marketing as a sequential decision problem with delayed effects and cumulative objectives such as customer lifetime value.

Contributions.

- We present a diagnostic evaluation pipeline for journey optimization that separates world-model training from policy grading to avoid circular evaluation.
- We propose a staged experimental program (randomized vs confounded logging, horizon and bias sweeps) that operationalizes five conceptual propositions about when world-model planning should help or fail.
- Using a Level-0 synthetic environment with delayed fatigue/churn effects, we empirically demonstrate a regime-dependent reality gap: under confounded logs, long-horizon planning becomes over-optimistic (simulation exceeds true performance), while randomized logs are more conservative on the analyzed subset.

2 From Response Prediction to Causal Decision Making

Traditional response models predict outcomes such as conversion or purchase. However, targeting customers with high predicted response may not create incremental value; those customers may have converted anyway.

2.1 Uplift modeling

Uplift modeling estimates incremental treatment effects. For binary treatment, the individual treatment effect is:

$$\tau_i = \mathbb{E}[Y^i(1) \mid X^i] - \mathbb{E}[Y^i(0) \mid X^i].$$

Uplift modeling thus represents a transition from prediction (“what will happen?”) to causal prediction (“what changes because of our intervention?”).

3 From Individual Interventions to Customer Journeys

Uplift modeling often operates at one decision point. Customer journeys add sequential dependence: the effect of an action depends on where it occurs in the journey and on latent state:

$$\text{Effect}(a_t \mid s_t).$$

The problem becomes sequential:

$$a_t^* = \arg \max_a \mathbb{E}[\text{future value} \mid s_t, a_t = a]$$

rather than maximizing immediate conversion.

4 Customer Journeys as Dynamic Systems

Customer journeys can be represented as dynamic systems. A marketer observes signals (purchases, site visits, exposures), while latent constructs (fatigue, loyalty, churn propensity) remain partially observed. This naturally motivates sequential state representation and transition modeling.

5 A Reinforcement Learning Perspective

An MDP formulation $M = (S, A, P, R, \gamma)$ formalizes sequential decisions. The objective:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^T \gamma^t r_t \right].$$

Marketing interpretation: state is customer state, action is intervention, reward encodes business value (revenue, margin, churn penalty, CLV), and policy is the decision strategy.

6 Model-Free versus Model-Based Decision-Making

Model-free approaches learn policies/values directly. Model-based approaches learn a model of dynamics, then plan within it. World models are a modern model-based instantiation: learn $P(s_{t+1} \mid s_t, a_t)$ (or latent equivalents), then plan using imagined rollouts.

7 World Models: Concept, Adaptation, and Latent State

A world model approximates environment dynamics sufficiently to support simulation/planning. In marketing, the true environment $P(s_{t+1} | s_t, a_t)$ is unknown; a learned approximation $\hat{P}(s_{t+1} | s_t, a_t)$ is fit from logged journeys. State may be explicit (engineered features) or latent (learned representations of history).

8 Stochasticity

Customer behavior is stochastic; a deterministic simulator is often inadequate. A probabilistic world model can represent uncertainty as part of the decision problem rather than an error term to be minimized away.

9 Planning with the World Model

Given a current state s_t , a planner evaluates action sequences over horizon H , simulates future states and rewards in $\hat{P}$, and chooses the best sequence by:

$$\hat{J} = \sum_{k=0}^H \gamma^k \hat{r}_{t+k}.$$

In practice, the system executes only the next action and replans (receding horizon / MPC).

10 Why Long-Term Value Matters

Marketing optimization often focuses on immediate metrics; customer-level optimization may target CLV. The optimal action under short-term objectives can differ from the optimal action under long-term objectives. A world model provides a mechanism to represent delayed effects and optimize cumulative value.

11 Relationship with Uplift Modeling

World models are not a replacement for uplift. Uplift estimates incremental effects at a decision point; world models represent state evolution after interventions and thereby support sequential optimization. Importantly, training a world model from observational logs does not resolve identification challenges; it shifts them into learning a simulator capable of credible counterfactual journeys.

12 The Counterfactual Problem

Sequential marketing is counterfactual: we observe one realized trajectory but need to compare alternative action sequences. The counterfactual space grows combinatorially with horizon. World models offer a mechanism to explore this space via simulation, but do not eliminate causal requirements.

13 The Problem of Model Error

A planner optimizes $J(\pi \mid \hat{P})$, not $J(\pi \mid P)$. If $\hat{P}$ is inaccurate, the resulting policy can be optimal in simulation but poor in the true environment. This risk grows with planning horizon as errors compound.

14 The Marketing World Model as a Decision Simulator

We summarize the closed-loop view:

Customer history $\rightarrow$ State representation $\rightarrow$ World model $\rightarrow$ Future simulation $\rightarrow$ Planning $\rightarrow$ Marketing action
with feedback: action $\rightarrow$ response $\rightarrow$ new state $\rightarrow$ repeat.

15 Related Work and Positioning

Simulator-based RL and world models have been explored in recommender systems and user engagement optimization. Synthetic retail environments have been proposed for benchmarking. LLM-based simulated users are emerging. This work emphasizes marketing-economics objectives and focuses on when world-model answers can be trusted, using known-ground-truth staged environments.

16 Research Questions

We organize the empirical program around:

- RQ1 — Dynamics.** Can a learned world model represent customer-state transitions following interventions?
- RQ2 — Optimization.** Can planning in the learned model improve cumulative value over non-sequential approaches?
- RQ3 — Long-term effects.** Does explicit modeling of state transitions produce different decisions than approaches optimized for immediate response?
- RQ4 — Planning horizon.** How does decision quality change as horizon increases?
- RQ5 — Model uncertainty.** How do uncertainty and prediction error affect policy reliability?

17 Conceptual Propositions

- P1.** If customer behavior has temporal dependencies, an explicit transition model should be more informative for sequential optimization than a single-step response model.
- P2.** The value of world-model planning should increase when actions have delayed/persistent effects.
- P3.** The advantage of long-horizon planning should decrease when prediction error becomes sufficiently large.
- P4.** Optimizing against an inaccurate world model can produce policies that perform well in simulation but poorly in the true environment (reality gap).
- P5.** World-model optimization should be most useful when the objective is cumulative (CLV) rather than purely immediate.

Table 1: Metrics mapping (as specified in the draft).

Proposition	Manipulated variable	Measured outcome
P1	Model class	Transition error vs response-model error
P2	Delayed-effect strength	Sequential vs myopic value gap
P3	Horizon H	Policy value vs H
P4	Injected bias β	Simulated value vs true value
P5	Discount factor γ	Advantage as $\gamma \rightarrow 1$

18 Proposed Research Framework

Customer history $\rightarrow$ State representation $\rightarrow$ World model $\rightarrow$ Future simulation $\rightarrow$ Planning $\rightarrow$ Marketing action, with feedback from response to state update. The core conceptual shift is from marketing prediction to marketing simulation.

19 Proposed Experimental Program

We specify a staged synthetic environment program with known ground-truth dynamics.

19.1 Environment: two levels of difficulty

Level 0 (primary). Fully observable state $s_t = (E_t, F_t, P_t) \in [0, 1]^3$ (engagement, fatigue, propensity). Actions $A = \{\text{none}, \text{email}, \text{discount}, \text{recommendation}\}$. Transitions follow a logistic-style update with noise and an engineered delayed cost: **discount** increases immediate purchase probability and increases future fatigue. Reward combines purchase revenue, discount cost, and churn/unsubscribe penalty when fatigue crosses a threshold.

Level 1 (robustness). Partially observable: a latent segment variable modulates coefficients; marketer observes only noisy proxies. This tests latent-state modeling claims.

19.2 Data generation

Trajectories are logged under (a) randomized policy (unconfounded coverage) and (b) confounded rule-based “business as usual” policy. World models are trained separately on each.

19.3 World model and planning

Model complexity is staged: shallow MLP for Level 0; recurrent/latent-state model for Level 1. Planning uses rollout-based MPC with discrete actions.

19.4 Baselines

Random policy; single-step / myopic uplift policy; world-model planner trained on randomized logs; planner trained on confounded logs; oracle planner using true dynamics; model-free RL with unlimited simulator access (ceiling).

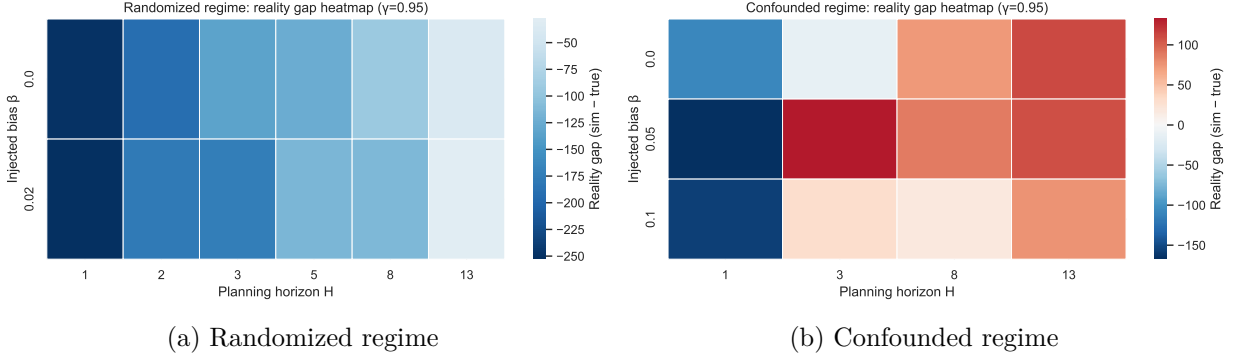


Figure 1: Reality gap heatmaps (simulated minus true return) for MPC as a function of horizon H and injected bias β (figure script fixes γ).

19.5 Metrics and proposition mapping

19.6 Observed Level-0 results (this work)

Using the implemented Level-0 environment and sweep pipeline, we directly measure Proposition P4 in the intended form: simulated policy value versus true policy value, and the resulting reality gap. We also compare randomized versus confounded logging regimes.

Key finding. Under randomized training logs (better coverage), simulated values are conservative in the analyzed subset (negative gap shrinking toward zero as H increases). Under confounded logs, long-horizon planning becomes optimistic (positive gap at larger H), matching the failure mode described in the simulator spec: planning exploits model error and the reality gap grows with horizon under biased data.

20 Expected Contributions

Conceptual. Marketing journey optimization is framed as a world-model-based sequential decision problem positioned against adjacent recommender-systems work that optimizes engagement rather than marketing economics.

Methodological. A staged, known-ground-truth environment and a causal-identification experiment (randomized vs confounded logging) are specified, with each proposition mapped to a runnable comparison.

Practical. The work aims to identify conditions—in terms of planning horizon and model error—under which simulated journeys provide useful decision support, and conditions under which they do not.

21 Limitations and Scope

A simulated environment cannot reproduce full real-world complexity. The validity of a learned world model depends on the quality and causal informativeness of the data. External factors (competition, macro conditions) are excluded. Reward reflects a chosen objective (revenue/margin/retention/CLV).

Predictive plausibility must be distinguished from causal validity. Simulation success does not imply safe deployment without additional validation and governance.

22 Discussion

The central conceptual shift is from marketing prediction to marketing simulation. World models add the question: “what might happen to the customer after I act?” This enables decision-time reasoning about consequences, but introduces a fundamental tension: longer-horizon planning increases both potential value and dependence on model accuracy. The research focus is therefore not whether a world-model marketing system can be built, but under what conditions its answers can be trusted.

23 Conclusion

This paper proposes learned world models as a framework for customer-journey and CLV optimization:

Customer state + Marketing action $\rightarrow$ Future customer state.

Once learned, the model can simulate alternative journeys and evaluate sequences of interventions, shifting the question from “which customer should receive this campaign?” to “which intervention sequence best achieves the desired long-term outcome?” The critical empirical chain is:

World-model accuracy $\rightarrow$ Simulation quality $\rightarrow$ Planning quality $\rightarrow$ Marketing performance.

Understanding this chain under known ground truth is essential to determine when world models represent meaningful marketing optimization versus predictive modeling with a more ambitious interface.

References

Table 2: Mean reality gap $\Delta = J(\pi \mid \hat{P}) - J(\pi \mid P)$ for MPC policies on a matched subset ($H \in \{1, 3, 8, 13\}$; $\gamma \in \{0.95, 0.99\}$). Positive values indicate optimistic simulation.

Regime	γ	β	H	Mean reality gap
Randomized (subset shown, $\beta = 0.0$)				
Randomized	0.95	0.00	1	-250.00
Randomized	0.95	0.00	3	-134.49
Randomized	0.95	0.00	8	-93.63
Randomized	0.95	0.00	13	-39.70
Randomized	0.99	0.00	1	-250.00
Randomized	0.99	0.00	3	-148.95
Randomized	0.99	0.00	8	-104.57
Randomized	0.99	0.00	13	10.78
Confounded				
Confounded	0.95	0.00	1	-108.60
Confounded	0.95	0.00	3	-13.98
Confounded	0.95	0.00	8	72.46
Confounded	0.95	0.00	13	110.12
Confounded	0.95	0.05	1	-167.42
Confounded	0.95	0.05	3	132.26
Confounded	0.95	0.05	8	85.60
Confounded	0.95	0.05	13	107.12
Confounded	0.95	0.10	1	-157.80
Confounded	0.95	0.10	3	29.86
Confounded	0.95	0.10	8	17.14
Confounded	0.95	0.10	13	74.84
Confounded	0.99	0.00	1	-108.60
Confounded	0.99	0.00	3	-9.00
Confounded	0.99	0.00	8	5.58
Confounded	0.99	0.00	13	99.34
Confounded	0.99	0.05	1	-167.42
Confounded	0.99	0.05	3	111.54
Confounded	0.99	0.05	8	62.58
Confounded	0.99	0.05	13	104.90
Confounded	0.99	0.10	1	-157.80
Confounded	0.99	0.10	3	-21.02
Confounded	0.99	0.10	8	95.04
Confounded	0.99	0.10	13	147.16